

Please check the examination details below before entering your candidate information			
Candidate surname		Other names	
Pearson Edexcel International Advanced Level	Centre Number		Candidate Number
Friday 17 January 2020			
Afternoon (Time: 1 hour 20 minutes)		Paper Reference WCH13/01	
Chemistry International Advanced Subsidiary/Advanced Level Unit 3: Practical Skills in Chemistry I			
Candidates must have: Scientific calculator Ruler			Total Marks

Instructions

- Use **black** ink or **black** ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*

Information

- The total mark for this paper is 50.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*
- You will be assessed on your ability to organise and present information, ideas, descriptions and arguments clearly and logically, including your use of grammar, punctuation and spelling.
- There is a Periodic Table on the back cover of this paper.

Advice

- Read each question carefully before you start to answer it.
- Show all your working in calculations and include units where appropriate.
- Check your answers if you have time at the end.

Turn over ►

P60471A

©2020 Pearson Education Ltd.

1/1/1/1/



Answer ALL the questions.

Write your answers in the spaces provided.

1 Tests were carried out on some pairs of compounds.

- (a) (i) Bromine water was added to separate solutions of sodium chloride and sodium iodide.

State **one** different observation for each reaction.

(2)

sodium chloride

.....

sodium iodide

.....

- (ii) Name a test, with the expected observation, to confirm the presence of the sodium ion in these compounds.

(2)

Test	Observation

- (b) (i) Barium chloride solution and hydrochloric acid were added to separate aqueous solutions of ammonium sulfate and ammonium nitrate.

State what would be **seen** for each compound which would allow you to distinguish between them.

(2)

ammonium sulfate

ammonium nitrate



- (ii) Give a test, with the expected result, to confirm the presence of the ammonium ion (NH_4^+) in the ammonium compounds.

(2)

Test	Result

- (c) (i) Acidified potassium dichromate(VI) solution was added to two test tubes each containing a different alcohol. The test tubes were placed in a warm water bath.

The alcohols were propan-1-ol and 2-methylpropan-2-ol.

State what would be **seen** for each alcohol which would allow you to distinguish between them.

(2)

propan-1-ol.....

2-methylpropan-2-ol.....

- (ii) Give a **chemical** test, with the expected observation, to confirm the presence of the hydroxy group.

(2)

Test	Observation

- (d) Acidified potassium manganate(VII) solution was added to separate test tubes containing samples of hexane and hexene. The test tubes were shaken gently.

State what would be **seen** for each compound which would allow you to distinguish between them.

(2)

hexane.....

hexene.....

(Total for Question 1 = 14 marks)

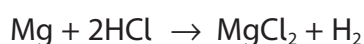


P 6 0 4 7 1 A 0 3 1 6

- 2 A class of students carried out experiments to determine the enthalpy change for the reaction of magnesium metal with hydrochloric acid.

The following method was used.

- Step 1 A 1.00 m length of magnesium ribbon was cleaned using sandpaper, weighed and cut into 10 cm lengths.
- Step 2 50 cm³ of dilute hydrochloric acid (an excess) was placed into a polystyrene cup and the temperature measured.
- Step 3 A 10 cm length of magnesium ribbon was added to the hydrochloric acid. The solution was stirred gently and the maximum temperature recorded.



Results

Measurement	Value
Mass of 1.00 m of magnesium ribbon / g	0.86
Initial temperature of hydrochloric acid before addition of magnesium ribbon / °C	21.4
Final temperature of solution / °C	29.2

- (a) (i) Calculate the number of moles of magnesium in the 10 cm length of ribbon used in this experiment. [*A_r* value: Mg = 24.3]

(2)



- (ii) Calculate the enthalpy change for this reaction including a sign and units. Give your answer to an appropriate number of significant figures.

Data:

Specific heat capacity of the solution = $4.2 \text{ J g}^{-1} \text{ }^{\circ}\text{C}^{-1}$

The density of the reaction mixture = 1.0 g cm^{-3}

(4)

- (b) (i) The maximum uncertainty each time the thermometer was read was $\pm 0.1 \text{ }^{\circ}\text{C}$. Calculate the percentage uncertainty in measuring the temperature change in this experiment.

(1)

- (ii) Suggest **one** way of reducing the percentage uncertainty in measuring the temperature change without changing the apparatus or just repeating the experiment. Justify your answer.

(2)

.....

.....

.....

.....



- (c) One student carried out the same experiment but used a glass beaker instead of a polystyrene cup.

State how this would affect the value of the enthalpy change obtained.
Justify your answer.

(2)

- (d) Explain why the magnesium ribbon was cleaned with sandpaper before being weighed.

(2)

(Total for Question 2 = 13 marks)



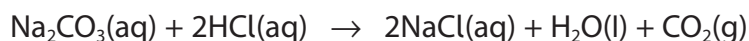
- 3 An experiment was carried out to determine the purity of solid sodium carbonate, Na_2CO_3 . The following procedure was used.

4.89 g of impure sodium carbonate was weighed and dissolved in distilled water.

The solution and washings were transferred to a 250.0 cm^3 volumetric flask, and the liquid level made up to the mark with distilled water and the flask shaken.

A pipette was used to transfer 25.0 cm^3 portions of the solution to conical flasks.

Each portion of the solution was then titrated with hydrochloric acid of concentration 0.200 mol dm^{-3} .



- (a) The indicator used was methyl orange. State the colour change at the end-point.

(2)

From to

- (b)

Results

Number of titration	1	2	3	4
Burette reading (final) / cm^3	27.55	26.25	28.30	26.15
Burette reading (start) / cm^3	0.00	0.05	1.05	0.05
Volume of $\text{HCl}(\text{aq})$ / cm^3				

- (i) Complete the table and, using appropriate titrations, calculate the mean titre.

(2)



(ii) Calculate the percentage purity, by mass, of the sodium carbonate.

(5)

(Total for Question 3 = 9 marks)



P 6 0 4 7 1 A 0 9 1 6

4 Bromoethane can be prepared by reacting ethanol with a mixture of sodium bromide and concentrated sulfuric acid.

- (a) **Step 1** 5 cm^3 of ethanol and 5 cm^3 of water are added to a round-bottomed flask. The flask is placed in an ice bath and 5 cm^3 of concentrated sulfuric acid is added slowly. During this process the flask is shaken gently.

Explain why the sulfuric acid must be added slowly.

(2)

- (b) **Step 2** 6.0 g of solid potassium bromide is ground up into a fine powder using a pestle and mortar. The powder is then added to the round-bottomed flask containing the ethanol and concentrated sulfuric acid. The mixture is heated.

State why the potassium bromide is ground up to a fine powder. Justify your answer.

(2)



(c) Step **3** The crude bromoethane formed in Step **2** is distilled off.

- (i) Draw a labelled diagram to show the apparatus suitable for this distillation.
Include a thermometer but no clamps or stands.

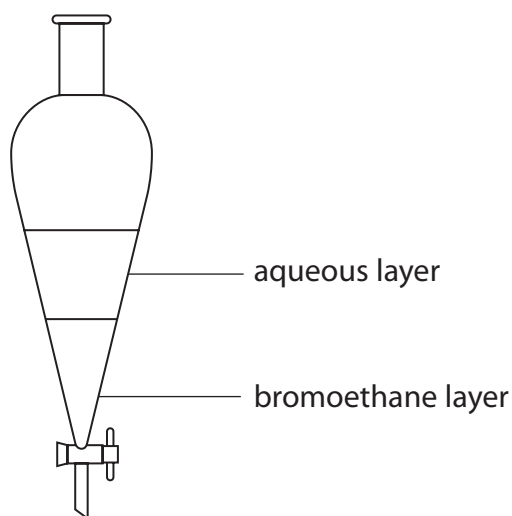
(3)



(ii) State how anti-bumping granules prevent bumping in the distillation flask.

(1)

- (d) Step 4 The distillate from Step 3 is transferred to a separating funnel where it separates into an aqueous layer and a layer containing impure bromoethane.



- (i) State **two** physical properties of bromoethane that can be deduced from this diagram.

(2)

- (ii) Describe how the aqueous layer could be removed from the separating funnel.

(1)



- (e) **Step 5** After removing the aqueous layer, sodium hydrogencarbonate solution is added to the impure bromoethane in a separating funnel and the two layers separated again.

State why sodium hydrogencarbonate solution is added to the impure bromoethane. (1)

- (f) **Step 6** The bromoethane is placed into a sample bottle and a drying agent is added.

(i) Identify, by name or formula, a suitable drying agent. (1)

(ii) Describe how the appearance of the bromoethane changes after the drying agent has been added and the mixture allowed to stand. (1)

(Total for Question 4 = 14 marks)

TOTAL FOR PAPER = 50 MARKS



The Periodic Table of Elements

1	2	3	4	5	6	7	0 (8)
1.0 H hydrogen 1							(18)
							4.0 He helium 2
Key							
(1)		(2)		(3)			
relative atomic mass atomic symbol name atomic (proton) number							
6.9 Li lithium 3	9.0 Be beryllium 4	10.8 B boron 5	12.0 C carbon 6	14.0 N nitrogen 7	16.0 O oxygen 8	19.0 F fluorine 9	20.2 Ne neon 10
23.0 Na sodium 11	24.3 Mg magnesium 12	27.0 Al aluminium 13	28.1 Si silicon 14	31.0 P phosphorus 15	32.1 S sulfur 16	35.5 Cl chlorine 17	39.9 Ar argon 18
39.1 K potassium 19	40.1 Ca calcium 20	69.7 Ga gallium 31	65.4 Zn zinc 30	63.5 Cu copper 29	58.7 Ni nickel 28	58.9 Co cobalt 27	55.8 Fe iron 26
85.5 Rb rubidium 37	87.6 Sr strontium 38	114.8 In indium 49	112.4 Cd cadmium 48	107.9 Ag silver 47	106.4 Pd palladium 46	102.9 Rh rhodium 45	101.1 Ru ruthenium 44
132.9 Cs caesium 55	137.3 Ba barium 56	204.4 Tl thallium 81	200.6 Hg mercury 80	197.0 Au gold 79	195.1 Pt platinum 78	192.2 Ir iridium 77	190.2 Os osmium 76
[223] Fr francium 87	[226] Ra radium 88	207.2 Pb lead 82	209.0 Bi bismuth 83	209.0 Po polonium 84	209.0 At astatine 85	210 Rn radon 86	[222] Xe xenon 54
Elements with atomic numbers 112-116 have been reported but not fully authenticated							
[112] Cn copernicium 112							
[113] Nh nihonium 113							
[114] Fl flerovium 114							
[115] Mc moscovium 115							
[116] Lv livermorium 116							
[117] Ts tennessine 117							
[118] Og oganesson 118							
[119] Uue unbinilium 119							
[120] Uub unbinilium 120							
[121] Uut untrium 121							
[122] Uuq unquadium 122							
[123] Uub unbinilium 123							
[124] Uuq unquadium 124							
[125] Uut untrium 125							
[126] Uuq unquadium 126							
[127] Uut untrium 127							
[128] Uuq unquadium 128							
[129] Uut untrium 129							
[130] Uuq unquadium 130							
[131] Uut untrium 131							
[132] Uuq unquadium 132							
[133] Uut untrium 133							
[134] Uuq unquadium 134							
[135] Uut untrium 135							
[136] Uuq unquadium 136							
[137] Uut untrium 137							
[138] Uuq unquadium 138							
[139] Uut untrium 139							
[140] Uuq unquadium 140							
[141] Uut untrium 141							
[142] Uuq unquadium 142							
[143] Uut untrium 143							
[144] Uuq unquadium 144							
[145] Uut untrium 145							
[146] Uuq unquadium 146							
[147] Uut untrium 147							
[148] Uuq unquadium 148							
[149] Uut untrium 149							
[150] Uuq unquadium 150							
[151] Uut untrium 151							
[152] Uuq unquadium 152							
[153] Uut untrium 153							
[154] Uuq unquadium 154							
[155] Uut untrium 155							
[156] Uuq unquadium 156							
[157] Uut untrium 157							
[158] Uuq unquadium 158							
[159] Uut untrium 159							
[160] Uuq unquadium 160							
[161] Uut untrium 161							
[162] Uuq unquadium 162							
[163] Uut untrium 163							
[164] Uuq unquadium 164							
[165] Uut untrium 165							
[166] Uuq unquadium 166							
[167] Uut untrium 167							
[168] Uuq unquadium 168							
[169] Uut untrium 169							
[170] Uuq unquadium 170							
[171] Uut untrium 171							
[172] Uuq unquadium 172							
[173] Uut untrium 173							
[174] Uuq unquadium 174							
[175] Uut untrium 175							
[176] Uuq unquadium 176							
[177] Uut untrium 177							
[178] Uuq unquadium 178							
[179] Uut untrium 179							
[180] Uuq unquadium 180							
[181] Uut untrium 181							
[182] Uuq unquadium 182							
[183] Uut untrium 183							

* Lanthanide series

* Actinide series

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

